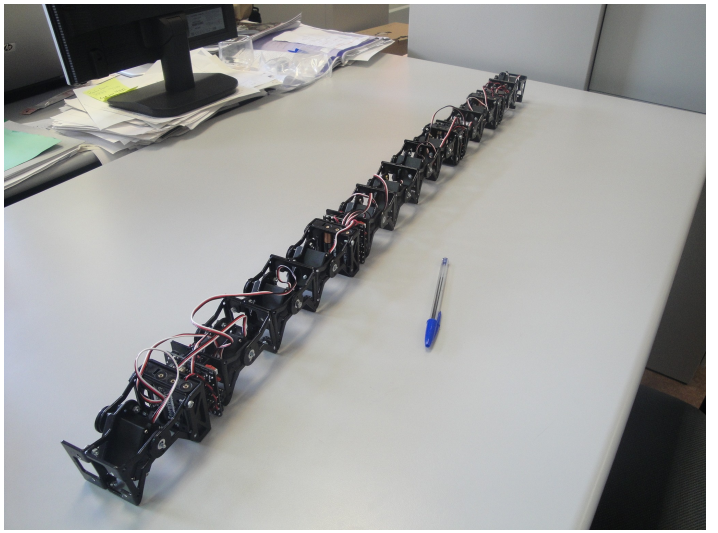
Toward the Sense of Touch in Snake Modular Robots for Search and Rescue Operations





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Outline

1. **Introduction**
2. Sense of touch
3. Mechanics
4. Experiments
5. Conclusions

Urban Search and Rescue Operations (I)

- Environment:
  - Unstructured terrain
  - Chaotic
  - Pieces of rumble
- Priority: to find survivals quickly
- Robot properties:
  - Versatility
  - Flexibility
  - Adaptability

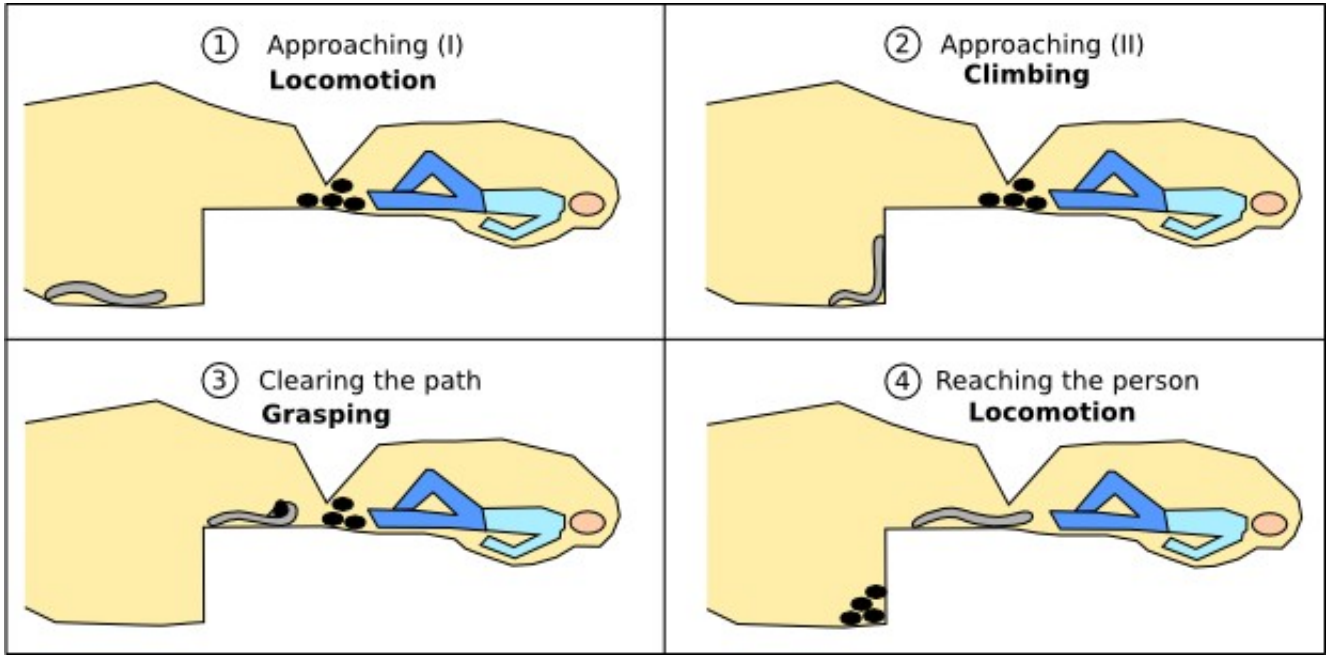


Urban Search and Rescue Operations (II)

- Snakes modular robots are very good candidates for search and rescue operations
  - Small section
  - Flexible body
  - Modular
    - Robustness
    - Versatile
    - Low cost

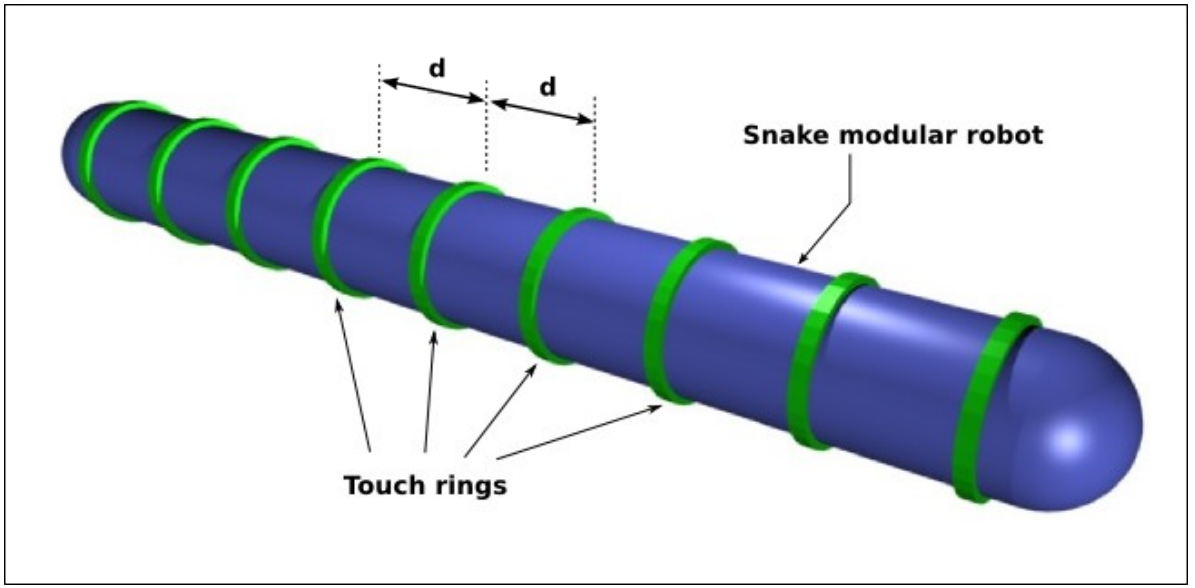


Urban Search and Rescue Operations (III)



• Capabilities needed (at least): Locomotion, climbing and grasping

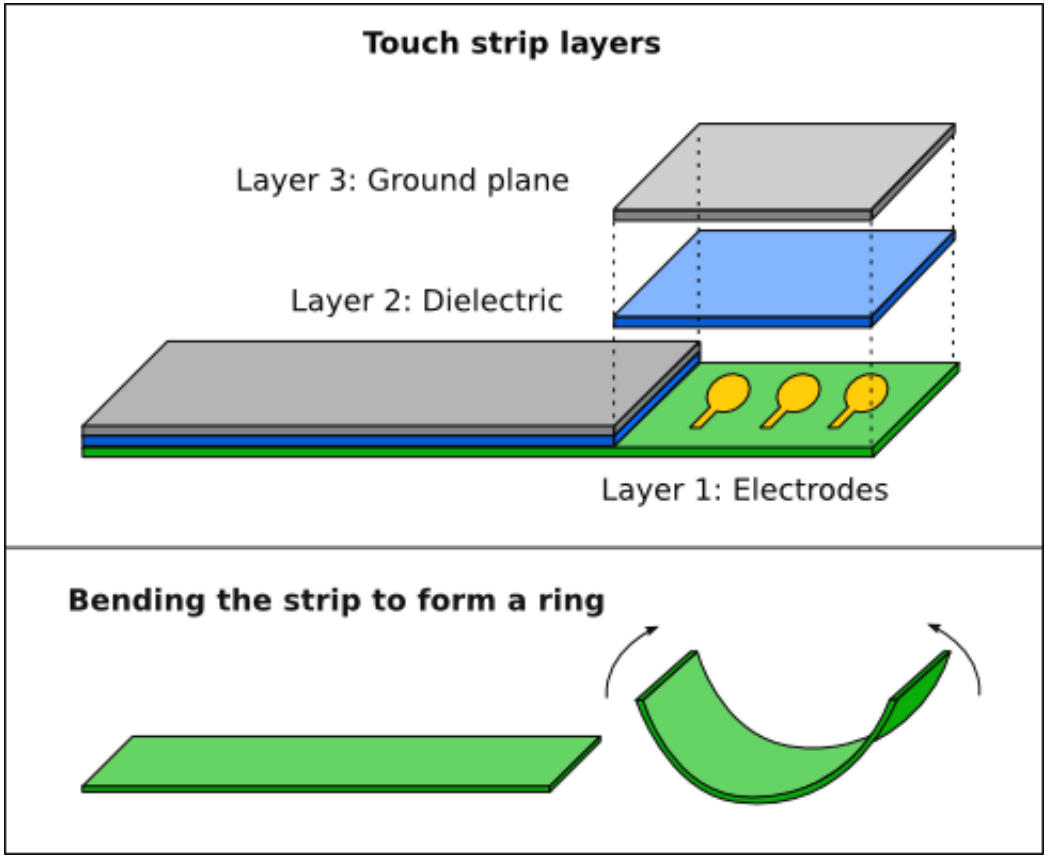
Touch rings



- Discrete touch rings vs continuous artificial skin
- Easier to adapt to the different snake prototypes
- Distance d can be changed according to the requirements

Touch strips

- Flexible capacitive strip
- Consist of three layers
- Electrodes in a flexible PCB layer
- The strip is bended to form a touch ring
- It fits snake robots with different cross section (circular, squared...)



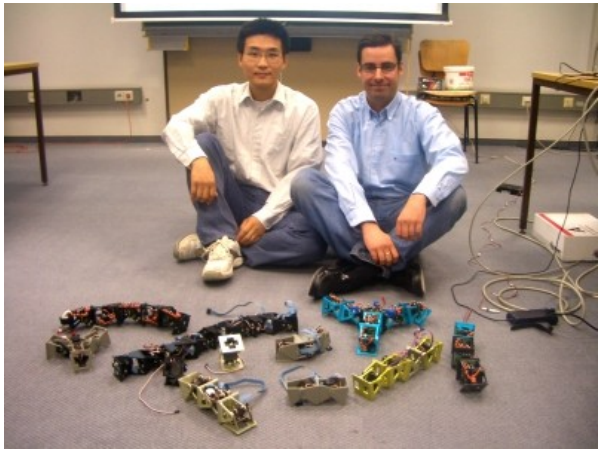
Modules

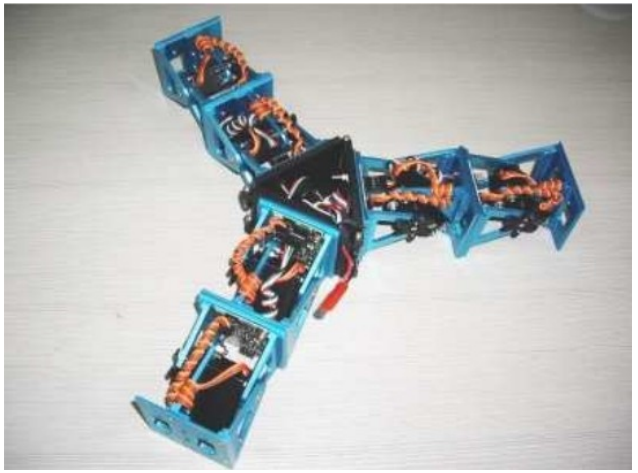


- Y1 modules family
- Cube-M family
- Joined together by screws

Modular snakes prototypes

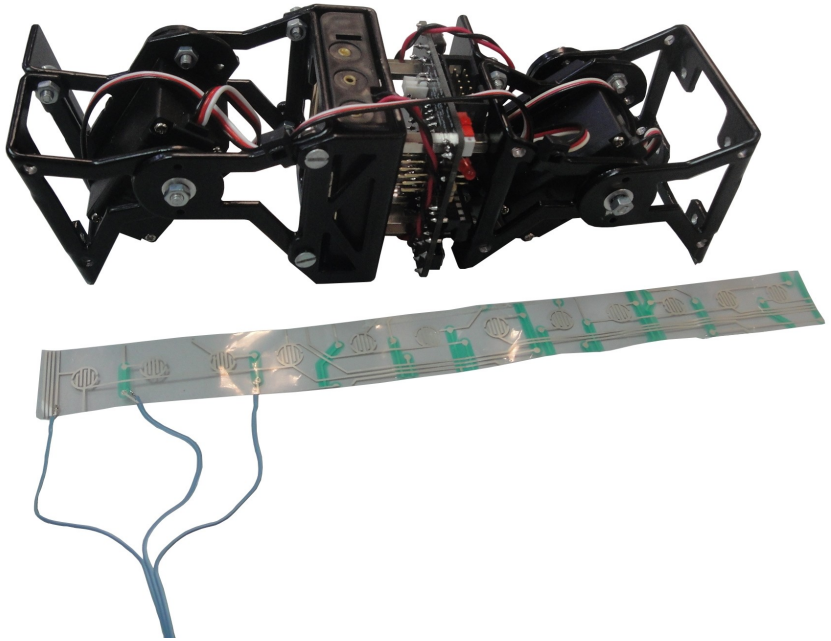


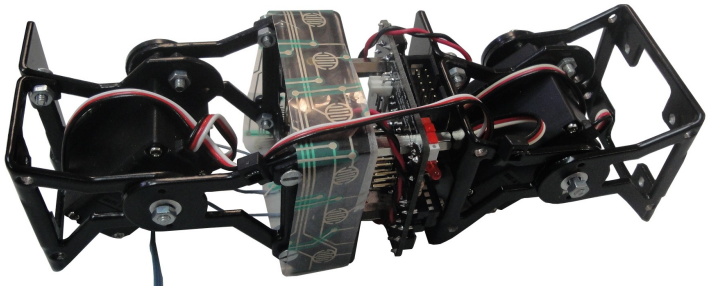






Touch strip experiments (I)

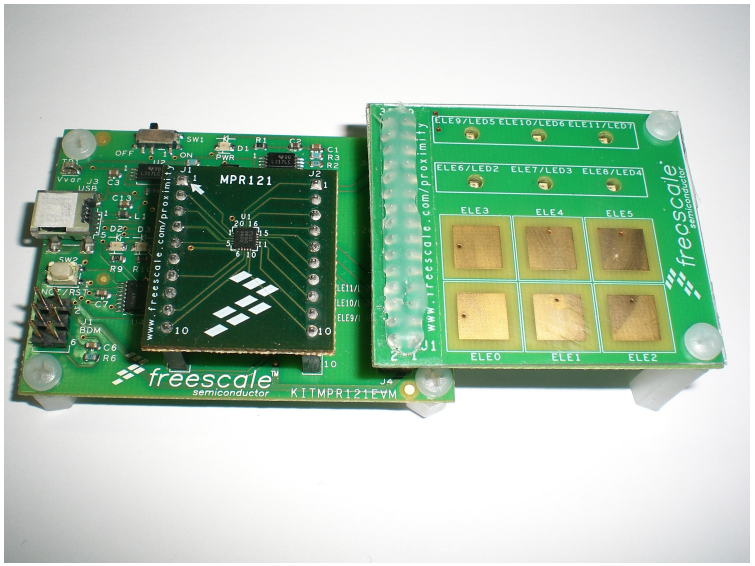


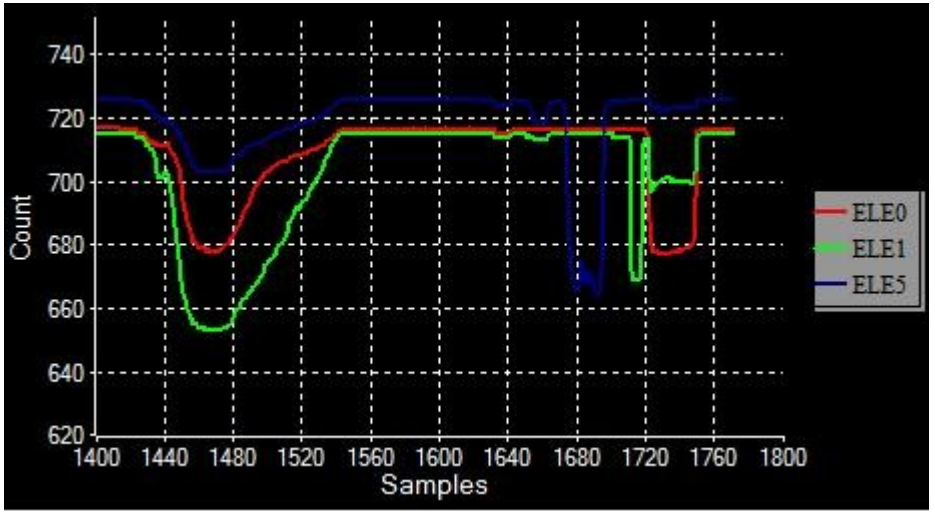


- Very early experiments
- Pith-pitch minimum modular robot

- Only the first touch strip layer
- Only three electrodes

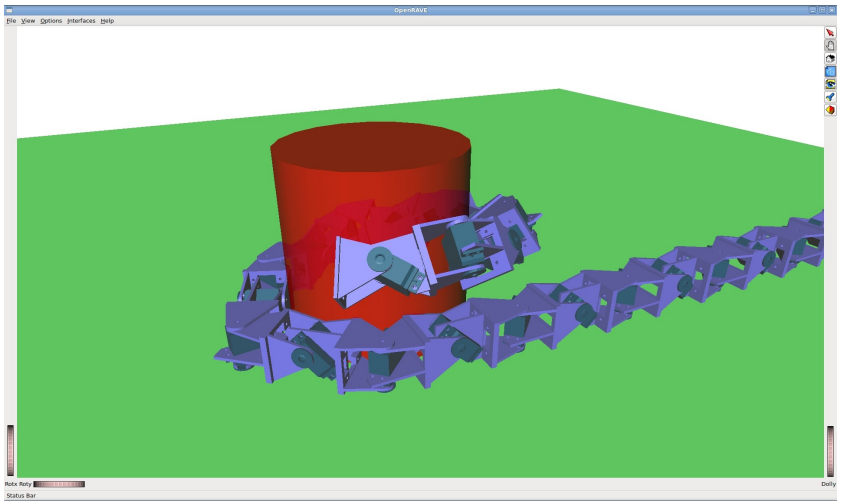
Touch strip experiments (II)

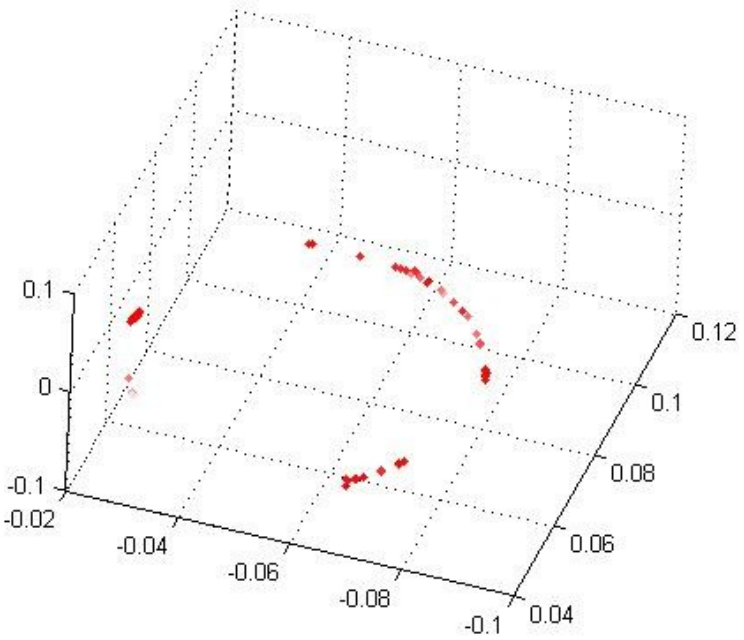




- First tests carried out with Freescale standard electrodes
- Substituted by our touch strip
- Freescale software is used to display the measures

Modular grasping simulation





• Grasping of a cylinder using the OpenRave Modular robot Plug-in (developed by ourselves)

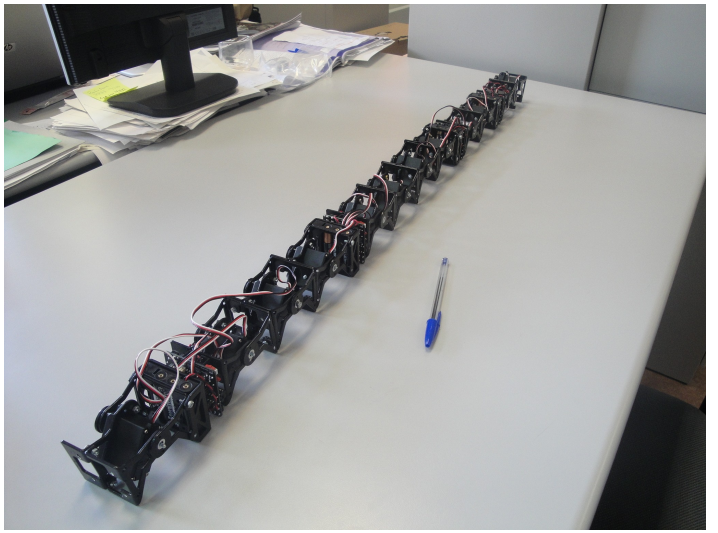
Conclusions

- Modular Snake robots are good candidates for being used in USAR Operations
- At least, locomotion, climbing and grasping capabilities are needed
- The sense of touch is key to achieving these capabilities
- Our idea is to use touch rings and touch strips
- The early experiments are promising….

…But still a lot of research have to be done

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